

Report on Finite-Time Formation Tracking Extension of the Swarm Kinematic Model

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1 Objective

This report summarizes the theoretical extension developed on top of the finite-time swarm tracking model of Fedele and D’Alfonso [1]. The objective is to introduce constant formation bias vectors p_i into the swarm dynamics such that the swarm can take any given shape while preserving the finite-time convergence properties of the original model.

The modified swarm dynamics are

$$\begin{aligned} \dot{x}_i(t) = & -(x_i(t) - p_i) + r(t) + \dot{r}(t) - \zeta(t) \\ & - \frac{M}{n} \sum_{j=1}^n \frac{(x_i(t) - p_i) - (x_j(t) - p_j)}{\|(x_i(t) - p_i) - (x_j(t) - p_j)\|^\nu + \varepsilon}, \end{aligned} \quad (1)$$

where $p_i \in \mathbb{R}^k$ denotes the desired offset of agent i with respect to the swarm centroid and $M \in \mathbb{R}^{d \times d}$ is a constant positive-definite interaction matrix governing the attraction–repulsion coupling, $\nu \in \mathbb{N}_+$ and $\varepsilon > 0$ are scalar tuning parameters, and the s -th component of $\zeta(t)$ is

$$\zeta_s(t) = \text{sign}(\sigma_s(t)) |\sigma_s(t)|^\beta, \quad \forall s = 1, \dots, d, \quad (2)$$

with $\beta \in (0, 1)$. The signal $\sigma(t)$ is the centroid tracking error,

$$\sigma(t) = c(t) - r(t), \quad c(t) = \frac{1}{n} \sum_{i=1}^n x_i(t), \quad (3)$$

where $c(t)$ is the swarm centroid and $r(t)$ is the reference trajectory to be tracked, $c(t) \rightarrow r(t)$.

Throughout this report the formation biases satisfy

$$\sum_{i=1}^n p_i = 0, \quad \dot{p}_i = 0, \quad \forall i \quad (4)$$

ensuring that the desired formation remains centered at the swarm centroid.

2 Centroid Convergence

Proposition 1. *Under the centering condition (4), the bias vectors p_i do not affect the centroid dynamics. Consequently, the swarm centroid converges to the reference trajectory $r(t)$ in finite time.*

Proof. Define

$$c(t) = \frac{1}{n} \sum_{i=1}^n x_i(t), \quad \sigma(t) = c(t) - r(t).$$

Differentiating gives

$$\dot{\sigma} = \frac{1}{n} \sum_{i=1}^n \dot{x}_i - \dot{r}.$$

Substituting the modified dynamics yields

$$\begin{aligned} \dot{\sigma} = & -c - \frac{M}{n^2} \sum_i \sum_j \frac{(x_i - p_i) - (x_j - p_j)}{\|(x_i - p_i) - (x_j - p_j)\|^\nu + \varepsilon} \\ & + r - \zeta. \end{aligned} \tag{5}$$

The interaction term remains antisymmetric,

$$f_{ij} = -f_{ji},$$

where

$$f_{ij} = \frac{(x_i - p_i) - (x_j - p_j)}{\|(x_i - p_i) - (x_j - p_j)\|^\nu + \varepsilon},$$

and therefore

$$\sum_i \sum_j f_{ij} = 0.$$

Hence

$$\dot{\sigma} = -\sigma - \zeta \tag{6}$$

This is exactly the centroid error dynamics obtained in Fedele and D'Alfonso [1]. Since the proof of finite-time convergence depends only on these error dynamics, the original Lyapunov argument applies directly. For completeness, the finite-time convergence of the centroid is summarized below.

Equation (6) represents k independent scalar subsystems,

$$\dot{\sigma}_j(t) = -\sigma_j(t) - \text{sign}(\sigma_j(t)) |\sigma_j(t)|^\beta, \quad j = 1, \dots, k.$$

Consider the Lyapunov candidate

$$V_j(t) = \frac{1}{2} \sigma_j^2(t).$$

Its derivative along the trajectories of the system is

$$\begin{aligned} \dot{V}_j(t) &= \sigma_j(t) \dot{\sigma}_j(t) \\ &= -\sigma_j^2(t) - |\sigma_j(t)|^{\beta+1} \\ &\leq -|\sigma_j(t)|^{\beta+1}. \end{aligned} \tag{7}$$

Since

$$|\sigma_j| = (2V_j)^{1/2},$$

it follows that

$$\dot{V}_j \leq -2^{(\beta+1)/2} V_j^{(\beta+1)/2} = -\alpha V_j^\eta,$$

where

$$\eta = \frac{\beta+1}{2}, \quad \alpha = 2^\eta.$$

Because $\beta \in (0, 1)$, we have $\eta < 1$, and standard finite-time stability theory yields

$$V_j^{1-\eta}(t) \leq V_j^{1-\eta}(t_0) - \alpha(1-\eta)(t-t_0),$$

for all $t \in [t_0, \tau_j]$, where the settling time is

$$\tau_j = t_0 + \frac{V_j^{1-\eta}(t_0)}{\alpha(1-\eta)}.$$

Hence

$$\sigma_j(t) = 0, \quad \forall t \geq \tau_j.$$

Since each coordinate converges independently, the centroid satisfies

$$c(t) = r(t), \quad \forall t \geq \tau,$$

where

$$\tau = \max_{j=1, \dots, k} \tau_j.$$

Therefore, the introduction of the constant formation bias preserves the finite-time centroid convergence established for the original swarm model. Finally, condition (4) ensures that the formation bias affects only the relative positions of the agents and not the centroid itself. $\square$

3 Formation Error Bounding

Proposition 2. *Assume the formation biases satisfy (4). Then the formation tracking error of each individual agent converges in finite time to a bounded neighborhood. Consequently, the modified dynamics provide a strict steady-state error bound ρ governing the maximum deviation of any agent from its desired formation slot.*

Proof. From the previous proposition, the centroid reaches the reference trajectory after finite time,

$$c(t) = r(t), \quad t \geq \tau_c.$$

Consequently, the global error term vanishes:

$$\zeta(t) = 0, \quad t \geq \tau_c.$$

The swarm dynamics become

$$\begin{aligned}\dot{x}_i &= -(x_i - p_i) + r + \dot{r} \\ &\quad - \frac{M}{n} \sum_j \frac{(x_i - p_i) - (x_j - p_j)}{\|(x_i - p_i) - (x_j - p_j)\|^\nu + \varepsilon}.\end{aligned}\tag{8}$$

Introduce the tracking error coordinate for agent i as the deviation from its target position:

$$\delta_i(t) = x_i(t) - p_i - r(t).$$

Since the formation biases are constant,

$$\dot{p}_i = 0,$$

and therefore

$$\dot{\delta}_i = \dot{x}_i - \dot{r}.$$

Substituting into the dynamics yields the explicit error dynamics:

$$\dot{\delta}_i = -\delta_i - \frac{M}{n} \sum_j \frac{\delta_i - \delta_j}{\|\delta_i - \delta_j\|^\nu + \varepsilon}.\tag{9}$$

This system is structurally identical to the aggregation dynamics analyzed in Theorem 2 of Fedele and D'Alfonso [1]. We now establish the error bound.

Consider the corresponding Lyapunov candidate for the individual tracking error:

$$\Delta_i = \frac{1}{2} \|\delta_i\|^2$$

The derivative of $\Delta_i(t)$ along the trajectories of the error dynamics (9) is

$$\begin{aligned}\dot{\Delta}_i(t) &= \delta_i^T(t) \dot{\delta}_i(t) \\ &= \delta_i^T(t) \left(-\delta_i(t) - \frac{M}{n} \sum_{j=1}^n \frac{\delta_i - \delta_j}{\|\delta_i - \delta_j\|^\nu + \varepsilon} \right) \\ &= -\|\delta_i\|^2 + \frac{1}{n} \sum_{j=1}^n \frac{\delta_i^T M (\delta_j - \delta_i)}{\|\delta_i - \delta_j\|^\nu + \varepsilon} \\ &\leq -\|\delta_i\|^2 + \frac{\|\delta_i\| \|M\|}{n} \sum_{j=1}^n \frac{\|\delta_j - \delta_i\|}{\|\delta_i - \delta_j\|^\nu + \varepsilon},\end{aligned}\tag{10}$$

where the last inequality follows from the Cauchy–Schwarz inequality.

Lemma 1. For $\varepsilon > 0$ and $\nu \in \mathbb{N}^+$,

$$h(f) = \frac{f}{f^\nu + \varepsilon}, \quad f \geq 0,$$

satisfies

$$h(f) \leq \frac{1}{\nu} \left(\frac{\nu-1}{\varepsilon} \right)^{(\nu-1)/\nu}.$$

Proof. For $\nu = 1$, $h(f) < 1$.

For $\nu > 1$, the stationary point satisfies

$$\bar{f} = \left(\frac{\varepsilon}{\nu-1} \right)^{1/\nu},$$

which is easily verified to be a global maximum since

$$\frac{d^2 h}{df^2}(\bar{f}) \leq 0.$$

Evaluating $h(\bar{f})$ gives the stated bound. $\square$

To establish the strict formation error bound, we define the maximum deviation radius ρ as:

$$\rho = \frac{\|M\|}{\nu} \left(\frac{\nu-1}{\varepsilon} \right)^{(\nu-1)/\nu} \quad (11)$$

and the associated invariant hyper-ball bounding the maximum allowable tracking error:

$$\mathcal{B} = \{\delta_i \in \mathbb{R}^k \mid \|\delta_i\| \leq \rho\} \quad (12)$$

Applying Lemma 1 to (10), the derivative of the Lyapunov function satisfies:

$$\dot{\Delta}_i \leq -\|\delta_i\|^2 + \frac{\|\delta_i\|^2}{\|\delta_i\|} \|M\| \frac{n-1}{n\nu} \left(\frac{\nu-1}{\varepsilon} \right)^{(\nu-1)/\nu}. \quad (13)$$

When the tracking error is outside the bounded hyper-ball $\mathcal{B}$, we have:

$$\|\delta_i\| \geq \rho,$$

which implies, by the definition of ρ , that:

$$\frac{1}{\|\delta_i\|} \leq \frac{\nu}{\|M\|} \left(\frac{\varepsilon}{\nu-1} \right)^{(\nu-1)/\nu}.$$

Substituting this into the inequality for $\dot{\Delta}_i$ yields:

$$\dot{\Delta}_i \leq -\|\delta_i\|^2 + \frac{n-1}{n} \|\delta_i\|^2 = -\frac{1}{n} \|\delta_i\|^2 = -\frac{2}{n} \Delta_i.$$

Integrating yields an exponential decay of the tracking error outside the bound:

$$\Delta_i(t) \leq \Delta_i(t_0) e^{-2(t-t_0)/n}.$$

Since entering the hyper-ball is equivalent to the condition:

$$\Delta_i \leq \frac{\rho^2}{2},$$

the corresponding finite entrance time satisfies:

$$\bar{t}_i = t_0 - \frac{n}{2} \log \left(\frac{\rho^2}{2\Delta_i(t_0)} \right).$$

Thus, every agent's tracking error δ_i enters the invariant set $\mathcal{B}$ in finite time and remains bounded by ρ thereafter.

Transforming back to the physical coordinates yields:

$$\|x_i(t) - p_i - r(t)\| \leq \rho, \quad \forall t \geq \max(\tau_c, \bar{t}_i).$$

Hence, the introduction of the constant formation bias successfully drives each agent to its target offset p_i around the reference $r(t)$. The steady-state formation error is strictly bounded by the radius ρ , guaranteeing stable, shape-preserving reference tracking for the swarm. $\square$

References

- [1] Giuseppe Fedele and Luigi D'Alfonso. A kinematic model for swarm finite-time trajectory tracking. *IEEE Transactions on Cybernetics*, 49(10):3806–3815, 2019.